

Hardware-Efficient Optimal Error-Priority Control for Series-End VSI With ZSC Suppression

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Abstract—This article presents a hardware-efficient optimal error-priority control (OEPIC) strategy for series-end voltage source inverters (SE-VSIs), enabling high-performance zero-sequence current (ZSC) suppression with minimal computational cost. Unlike conventional methods such as multivector MPC or 3-D-space vector modulation (SVM), OEPIC eliminates complex real-time calculations and auxiliary compensators by using a compact 1-D lookup table (LUT) for direct phase-wise current control in ABC coordinates. This deterministic LUT-based approach achieves sub-microsecond execution on low-cost digital signal processor (DSPs) and is easily deployable on microcontrollers or programmable logic devices, making it ideal for cost-sensitive industrial applications. In addition, OEPIC integrates adaptive switching frequency control to reduce switching losses and improve electromagnetic interference (EMI) performance. Comprehensive validation through digital simulations, controller-hardware-in-the-loop (C-HIL) tests, and laboratory experiments with physically unbalanced RL loads demonstrate robust ZSC suppression, superior current quality, and maximal DC-bus utilization under challenging conditions. A comparative analysis further confirms OEPIC's intrinsic fault tolerance against inter-turn short circuits (ITSCs), preventing the current runaway observed in conventional schemes. These results establish OEPIC as a practical, scalable alternative to computationally intensive modulation-based strategies for industrial motor drives.

Index Terms—Field-oriented control (FOC), lookup table-based control, PMSM, series-end voltage source inverter (SE-VSI), zero-sequence current (ZSC).

I. INTRODUCTION

PERMANENT magnet synchronous motors (PMSMs), particularly open-winding (OW) configurations, are gaining attention for their high efficiency and fault tolerance [1], [2], [3], [4], [5], [6], [7], [8]. To further optimize these drives, series-end voltage source inverters (SE-VSIs) have emerged as

a cost-effective alternative to dual-inverter topologies, offering comparable DC-link utilization with fewer switching devices [2], [3], [9]. However, the absence of a neutral point in SE-voltage source inverter (VSI)s introduces zero-sequence current (ZSC) paths. Uncontrolled ZSC degrades performance by increasing harmonic distortion, torque ripple, and EMI [3], [4], [5], [6], [10].

Various strategies have been proposed to mitigate ZSC. Carrier-based pulse-width modulation (PWM) (CBPWM) methods limit common mode voltage (CMV) but often introduce switching harmonics and fail to eliminate dead-time induced spikes [3]. Advanced solutions like model predictive current control (MPCC) [4], [5], [11] and three-dimensional space vector modulation (3-D-SVM) [6] offer superior dynamic performance and overmodulation capabilities. However, 3-D-SVM requires complex sector identification and dwell-time calculations [6], while MPCC relies on iterative optimization models that are sensitive to parameter mismatches [4], [5]. Recent neural network-based approaches [7] further increase computational complexity. In addition, standard solutions often require separate linear controllers [e.g., proportional-integral-resonant (PIR) compensators] [14], [15] or struggle with high switching losses [12], [13], necessitating high-performance DSPs for implementation [8].

While modern DSPs can handle these intensive algorithms, doing so consumes critical processing resources required for other tasks, such as multiaxis synchronization or industrial communication, and increases hardware costs. Consequently, there is a distinct need for hardware-efficient control schemes that deliver high-quality ZSC suppression, even in the presence of non-sinusoidal back-electromotive force (EMF) [2], without incurring the heavy arithmetic burden of modulation synthesis or iterative optimization.

To address these limitations, a novel optimal error-priority control (OEPIC) strategy is proposed. OEPIC is a deterministic logic-based control strategy that achieves instantaneous multi-objective optimality per control cycle by replacing complex 3-D modulation synthesis and iterative optimization with a compact LUT for direct phase-wise current control.

Key contributions of OEPIC include.

- 1) *A Logic-Based Control Framework*: Direct ZSC suppression in ABC coordinates using a compact 1-D-LUT, eliminating the need for duty cycle averaging, coordinate transformations, or additional linear compensators.
- 2) *Hardware Efficiency*: Sub-2 μ s execution time on a standard TMS320F28335 microcontroller unit (MCU).

Received 12 June 2025; revised 29 December 2025, 1 June 2026, and 22 July 2026; accepted 30 July 2026. (Corresponding author: Eli Barbie.)

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Digital Object Identifier 10.1109/TIE.2026.3720735

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The logic-based architecture allows implementation on low-cost microcontrollers or simple programmable logic arrays (PLAs), reducing system cost.

- 3) *Power Efficiency*: Dynamic adjustment of the switching frequency (f_{sw}) to correct errors only when necessary, reducing switching losses by up to $\sim 30\%$ compared with fixed-frequency PWM.
- 4) *Comprehensive Validation*: Verified through digital simulations, C-HIL, and a laboratory SE-VSI prototype, demonstrating robustness against severe parameter asymmetry ($\pm 25\%$) and internal machine faults (inter-turn shorts), showing superior dynamic stability compared with standard CBPWM, while outperforming 3-D-SVM in terms of computational burden.

Moreover, regarding scalability, OEPC's adaptability could extend to other motor drive configurations, including polyphase SE-VSI-fed PMSMs and induction motors, provided the switching state (SS) impact table is derived for the specific topology. This underscores OEPC's industrial relevance as a high-performance, computationally efficient alternative.

The remainder of this article is structured as follows. Section II outlines the SE-VSI topology and the specific control challenges it presents. Section III details the OEPC methodology, error prioritization, and LUT design. Section IV presents comprehensive validation results, including comparative simulations and new experimental results from an unbalanced RL load prototype. Section V concludes the work.

II. BACKGROUND AND FORMULATIONS

A. OW-PMSM Model With ZSC Considerations

ZSC in OW-PMSMs arises primarily from two mechanisms: 1) inverter-side CMV resulting from sub-optimal modulation sequences and VSI nonlinearities, such as deadtime and device voltage drops [2], [4]; and 2) motor-side CMV due to third-harmonic rotor flux, which leads to non-sinusoidal back-EMF, particularly in salient pole interior PMSMs [11], [15]. In addition, resolver excitation frequencies of some speed encoders may also contribute to ZSC [1]. To account for these ZSC sources, the voltage-behind-reactance (VBR) model [16] is enhanced to incorporate third-harmonic flux effects [2], [17]. The resulting system, governed by (1)–(6), provides the necessary fidelity for the simulation and control design

$$\begin{cases} \lambda_d = L_d i_d + \lambda_m, & \lambda_q = L_q i_q \\ \lambda_0 = L_0 i_0 + K_m \lambda_m \cos(3\omega_e t) \end{cases} \quad (1)$$

$$\begin{cases} v_d \approx r_s i_d + e_d \\ v_q \approx r_s i_q + e_q \\ v_0 \approx r_s i_0 + L_0 \frac{di_0}{dt} + e_0 \end{cases} \quad (2)$$

$$\begin{cases} e_d \approx (L_d - L_q) \omega_e i_q \\ e_q \approx (L_d - L_q) \omega_e i_d + \omega_e \lambda_m \\ e_0 \approx -3K_m \lambda_m \omega_e \sin(3\omega_e t) \end{cases} \quad (3)$$

$$J \frac{d\omega_m}{dt} + F \omega_m = T_e - T_m \quad (4)$$

$$T_e = \frac{3p}{2} ((L_d - L_q) i_d i_q + \lambda_m i_q) - 9p K_m \lambda_m i_0 \sin(3\theta_e) \quad (5)$$

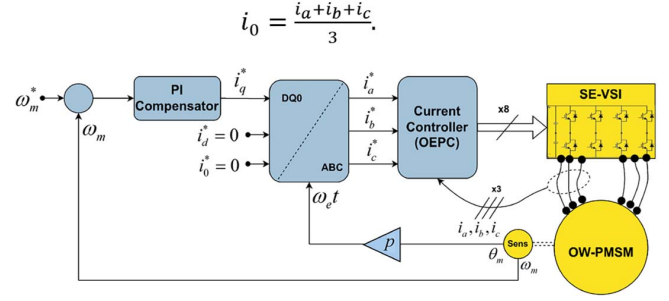


Fig. 1. Block diagram of OW-PMSM FOC-based speed control with ABC coordinates current controller (OEPC).

where standard notation applies for pole pairs (p), flux linkages (λ), voltages (v), back-EMF (e), currents (i), and torque (T). The subscript m denotes mechanical variables, while e denotes electrical quantities. K_m represents the ratio of the third-harmonic flux to the fundamental PM flux (λ_m). The ZSC is defined as

$$i_0 = \frac{i_a + i_b + i_c}{3}. \quad (6)$$

Note that the steady state back EMF expressions are obtained by transforming the currents components into their stationary reference frame (SRF) to facilitate phasor-based derivative approximations ($di_d/dt \approx +\omega_e i_q$ and $di_q/dt \approx -\omega_e i_d$).

B. OW-PMSM Field Oriented Control

The drive utilizes a field-oriented control (FOC) structure, as shown in Fig. 1. A PI-based speed controller generates the q-axis current reference (i_q^*) from the speed error, while the d-axis (i_d^*) and zero-sequence (i_0^*) references are maintained at zero to minimize losses and torque ripple. The speed loop PI gains are tuned using the symmetric optimum method [18] to achieve the desired bandwidth, while the proposed OEPC algorithm replaces the conventional inner current loop regulators.

C. SE-VSI Topology and Control Limitations

The SE-VSI topology (Fig. 2) comprises four independent half-bridge legs (L_1 – L_4) and operates without a neutral point. This structure produces $2^4 = 16$ feasible SS. Table I summarizes the effect of these states on the phase currents (i_a, i_b, i_c) and the ZSC (i_0). For clarity, switching trajectories are expressed as 4-bit binary sequences (L_1, L_2, L_3, L_4), where each bit represents the gate command of a corresponding leg: a value of 1 indicates the upper device is ON (lower OFF), while 0 indicates the lower device is ON (upper OFF). The table also captures the instantaneous influence of each SS on the currents. A preceding minus sign (–) denotes a decrease, whereas a plus sign (+) indicates an increase. The zero vectors (#0 and 15) do not affect the currents and are therefore unused. Notably, 8 out of the 16 SS impact the ZSC, while 6 SS impact each of the three-phase currents (with overlaps). This distribution creates a complex control scenario where multiple objectives must be balanced simultaneously.

The SS mapping summarized in Table I describes the instantaneous influence of each SS on the current magnitude and

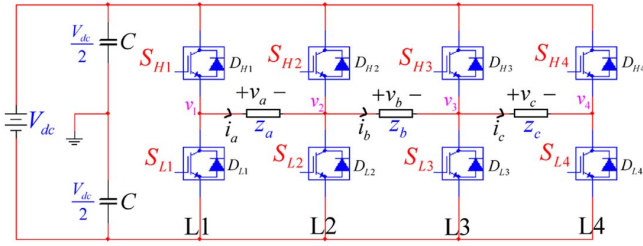


Fig. 2. Three-phase SE-End-VSI with open-winding load.

TABLE I
SWITCHING STATE IMPACT TABLE

#	SS($L_1 L_2 L_3 L_4$)	Impact
0	0000	N/A
1	0001	$-i_c, -i_0$
2	0010	$-i_b, +i_c$
3	0011	$-i_b, -i_0$
4	0100	$-i_a, +i_b$
5	0101	$-i_a, +i_b, -i_c, -i_0$
6	0110	$-i_a, +i_c$
7	0111	$-i_a, -i_0$
8	1000	$+i_a, +i_0$
9	1001	$+i_a, -i_c$
10	1010	$+i_a, -i_b, +i_c, +i_0$
11	1011	$+i_a, -i_b$
12	1100	$+i_b, +i_0$
13	1101	$+i_b, -i_c$
14	1110	$+i_c, +i_0$
15	1111	N/A

steering behavior of the phase currents, which governs their evolution over a control interval. Clearly, due to the inherent continuity requirement of inductive loads, the phase currents through the load impedances Z_{abc} (cf. Fig. 2) remain uninterrupted. A leg's binary state of 0 or 1 clamps its terminal to the negative or positive DC rail, respectively, but still allowing bidirectional current flow (either through the lower or upper transistors, or through their antiparallel diode). This ensures smooth and continuous operation across all conditions.

The ZSC component flows along a path that consists of the VSI's CMV, defined as the outer legs' voltage difference, $v_1 - v_4$ (c.f. Fig. 2), and a pseudo-series connection of the load impedances [6, Fig. 2(b)]. Many existing SE-VSI control schemes utilize this concept to suppress ZSC by controlling the SE-VSI's CMV [2], [3]. However, a critical control challenge in this topology is the presence of antiparallel diodes, which prevent the load impedances from forming simple series connections during off-states. Circuit analysis reveals that when a ZSC-impacting SS is applied, the antiparallel diodes force the load impedances into parallel or self-circulating paths, limiting the net ZSC impact to a 1/3-fold effect. Consequently, the SE-VSI can effectively control only one single-phase current at a time. This physical constraint renders standard modulation techniques suboptimal and motivates the proposed OEPC framework, which leverages error prioritization to handle this sequential control limitation seamlessly.

III. OEPC

The proposed OEPC framework addresses the “one-current-at-a-time” ZSC control limitation by employing a 1-D-LUT for optimal SS selection, integrating three-phase control and ZSC suppression into a seamless control process by utilizing error priority-based criteria to optimize the phase-wise current control. This efficient ON-OFF-based current control results in a variable f_{sw} , bounded by an adjustable upper limit of f_{max} , which is set by the sampling period, T_s , and given by the following relation [19], [20]:

$$f_{max} = \frac{1}{2T_s}. \quad (7)$$

Implemented directly in the SRF, the OEPC process eliminates the need for 3-D-SVM averaging and additional linear compensators while ensuring minimal switching losses through its efficient self-adjusted f_{sw} , which varies between $0.5f_{max}$ and f_{max} , with an average of $0.75f_{max}$.

A. Differential Mode and Common Mode Current Errors

With the employed phase-wise current regulation, each phase current (i_x , $x \in \{a, b, c\}$) can correct either a differential-mode (DM) error (for torque control) or a common-mode (CM) error (for ZSC suppression). This yields six independent errors: three DM and three CM errors (one per phase). The DM reference currents exclude the desired ZSC reference, i_0^* , which is ideally zero. Namely, for phase x , with complimentary phases y, z (e.g., $x = a \Rightarrow y = b, z = c$), there is

$$i_{DM(x)}^* = i_x^* - i_0^* = \frac{2i_x^* - i_y^* - i_z^*}{3}, \text{ where } i_0^* = \frac{i_x^* + i_y^* + i_z^*}{3}$$

and the three DM tracking errors become

$$\varepsilon_{DM(x)} = i_{DM(x)}^* - i_x = \frac{2i_x^* - i_y^* - i_z^*}{3} - i_x \quad (8)$$

where the $*$ superscript denotes a reference value. By equating i_0^* with i_0 (6), then solving for i_x , there is: $i_x = 3i_0^* - (i_y + i_z)$, where the right side of the equality serves as the reference value. As a result, the three CM errors become

$$\varepsilon_{CM(x)} = \frac{3i_0^* - (i_y + i_z) - i_x}{3}. \quad (9)$$

A 1/3 scaling factor was applied to balance the correction priority, avoiding the CM error corrections from disproportionately overshadowing the DM current regulation.

B. Error-Priority Selection and Reduction

With two error types per phase, where each phase current can be governed to regulate one error type at a time (DM or CM), an error-type priority selection process must be implemented by applying a pair-wise error comparison, in which the absolute values of DM and CM errors are compared per-phase. This way, the most critical phase-wise error is selected, thus reducing the error set from six to three (one critical per phase), viz

$$\varepsilon_{crit(x)} = \varepsilon_{DM(x)} \text{ if } |\varepsilon_{DM(x)}| \geq |\varepsilon_{CM(x)}| \text{ else } \varepsilon_{CM(x)}. \quad (10)$$

TABLE II
ERROR PRIORITY CODE
and ORDER

#	EPC($P_{ab}P_{bc}P_{ca}$)	EPO
0	000	N/A
1	001	CBA
2	010	BAC
3	011	BCA
4	100	ACB
5	101	CAB
6	110	ABC
7	111	N/A

In addition, an error mode predominance (EMP) flag, F_{mp} , is set to 0 if the majority of the phase-wise critical errors (CEs) are of CM type. Else, it is set to 1, viz

$$F_{mp} = \begin{cases} 1, & \text{if } 2 \varepsilon_{crit(x)} \text{ are of DM type} \\ 0, & \text{if } 2 \varepsilon_{crit(x)} \text{ are of CM type.} \end{cases} \quad (11)$$

Next, the three selected phase-wise CEs, $\varepsilon_{crit(x)}$, are sorted in descending order of absolute values, while their order is encoded as a 3-bit error priority code (EPC) defining which phase-wise error is maximum (highest priority), intermediate (medium priority), and minimum (lowest priority). The EPC, which consists of three binary variables, P_{ab} , P_{bc} , and P_{ca} , is obtained using three logical comparisons

$$\begin{cases} P_{ab} = [|\varepsilon_{crit(a)}| > |\varepsilon_{crit(b)}|] \\ P_{bc} = [|\varepsilon_{crit(b)}| > |\varepsilon_{crit(c)}|] \\ P_{ca} = [|\varepsilon_{crit(c)}| > |\varepsilon_{crit(a)}|] \end{cases} \quad (12)$$

where $[]$ is the Iverson brackets notation. The error priority orders (EPOs) and their corresponding EPCs are listed in Table II, where an EPO of CBA (EPC = 001) means that $\varepsilon_{crit(c)}$ is the most critical (the prime error), followed by phases B and A, namely, $|\varepsilon_{crit(c)}| > |\varepsilon_{crit(b)}| > |\varepsilon_{crit(a)}|$. Note that 000 and 111 are unused since only $3! = 6$ unique permutations exist. In addition to the EPCs, the sorting process also generates a 3-bit error signs code (ESC), indicating each CE's sign using a second set of binary variables, S_{mx} , S_{md} , and S_{mn} , corresponding to the highest priority, medium priority, and least priority errors, respectively, where a value of 0 (negative) means that the current must be decreased and a value of or 1 (positive) means that the current must be increased.

For example, if EPC = 001 (CBA priority) and ESC = 101, then the most CE, $\varepsilon_{crit(c)}$, is positive, the intermediate CE, $\varepsilon_{crit(b)}$, is negative, and the least CE, $\varepsilon_{crit(a)}$, is positive. The three ESC bits can be logically expressed as follows:

with: $S_x = [\varepsilon_{crit(x)} > 0]$, and $x \in \{a, b, c\}$,

$$\begin{cases} S_{mx} = S_a \overline{P_{ca}} P_{ab} + S_b \overline{P_{ab}} P_{bc} + S_c \overline{P_{bc}} P_{ca} \\ S_{md} = S_a (\overline{P_{ca}} \oplus \overline{P_{ab}}) + S_b (\overline{P_{ab}} \oplus \overline{P_{bc}}) + S_c (\overline{P_{bc}} \oplus \overline{P_{ca}}) \\ S_{mn} = S_a P_{ca} \overline{P_{ab}} + S_b P_{ab} \overline{P_{bc}} + S_c P_{bc} \overline{P_{ca}} \end{cases} \quad (13)$$

The seven key binary variables (7-bit LUT input address)

$$[P_{ab}, P_{bc}, P_{ca}, S_{mx}, S_{md}, S_{mn}, F_{mp}] \quad (14)$$

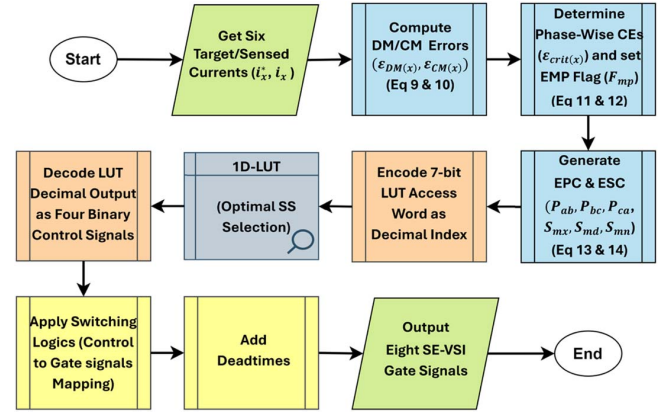


Fig. 3. Overall implementation flowchart of the main OEPC control framework with 1-D-LUT-based optimal switching state selection.

are encoded into a 7-bit decimal index. The 1-D-LUT then utilizes this index to determine the optimal SSs for the next processing step.

C. Switching State Selection (LUT Programming)

The core of OEPC lies in its comprehensive decision-making, which follows a structured, precomputed mapping rather than a runtime-based, computationally intensive algorithm for the optimal selection of SSs. This is achieved through a preprogrammed 1-D-LUT that maps the 7-bit input address (14), supplied as a pre-encoded decimal index, directly to the most appropriate SS. The 1-D-LUT has 96 entries, derived from: $3! = 6$ EPC permutations, $2^3 = 8$ ESC combinations, and $2^1 = 2$ EMP states, resulting in $3 \cdot 8 \cdot 2 = 96$ total entries, where the remaining 32 entries of the potential $2^7 = 128$ LUT input addresses (14) are unused. Each entry corresponds to one of the 14 valid non-zero SSs (cf. Table I). Since the SE-VSI topology inherently links adjacent legs with phase currents, thus preventing simultaneous independent correction of multiple currents (cf. Section II), a strategic error-prioritized decision-making process is utilized. The SS selection process depends on the computed CEs and their characteristics, which are fully represented by the 7-bit input address (14). The LUT outputs the four switching functions of the SE-VSI, S_1, S_2, S_3, S_4 as an encoded decimal variable, which can be decoded into the eight SE-VSI's gate signals (cf. Figs. 1 and 3). Due to the phase/leg interdependence, if one current is increased/decreased, any adjacent current can either be decreased/increased or preserved. For example, setting $S_1 = 1$ and $S_2 = 0$ applies $+V_{DC}$ to phase-A, increasing i_a . This leaves two options for governing i_b , either setting $S_3 = 1$ to decrease i_b by applying $-V_{DC}$ to phase-B, or setting $S_3 = 0$, preserving i_b through self-circulation by applying zero voltage to phase-B (cf. Fig. 1 and Table I).

The SS selection process follows four fundamental rules that govern the LUT design.

- 1) *Prime Error Correction Priority*: The most critical error (prime error) as determined by the EPC is always corrected and never preserved. This ensures that the most significant control deviation receives immediate attention.

TABLE III
SELECTED OPTIMAL SS LUT ENTRIES

Case#	EPO	EPC	ESC	EMP	SS	Impact
1	ACB	100	011	1	0110	$-i_a, +i_c$
2	ACB	100	001	0	0101	$-i_a, +i_b, -i_c, -i_0$
3	BAC	010	110	1	1101	$+i_b, -i_c$
4	BAC	010	110	0	1100	$+i_b, +i_0$
5	ABC	110	001	1	0110	$-i_a, +i_c$
6	ABC	110	001	0	0111	$-i_a, -i_0$
7	CAB	101	010	1	1001	$+i_a, -i_c$
8	CAB	101	110	0	1010	$+i_a, -i_b, +i_c, +i_0$

- 2) *Maximum Error Correction*: The selected SS attempts to correct as many CEs as possible. Secondary CE corrections are applied if they can be addressed without compromising the prime error correction.
- 3) *Non-Interference Principle*: Any secondary CE that cannot be immediately corrected is preserved through self-circulation of its corresponding current, thus ensuring a “do no harm” policy.
- 4) *Error Mode Predominance Selection*: When $F_{mp} = 1$ (DM predominance), the candidate SS subset consists of the 6 SSs that do not impact i_0 . Conversely, when $F_{mp} = 0$ (CM predominance), the candidate subset includes the 8 SSs that directly impact i_0 . This ensures optimal balancing between phase current regulation and ZSC suppression.

Since the full 96-entry LUT is too large to be listed entirely, Table III lists eight key LUT entries, while the full LUT is provided in the online supplementary material [21].

It is worth clarifying that the term “optimal” is used here in an instantaneous finite-state lexicographic sense. For each control cycle, the SE-VSI provides a finite set of admissible nonzero SSs, whose current impacts are completely defined by Table I. OEPC selects the SS satisfying the lexicographic priority: first enforcing correction of the prime error, then maximizing the number of corrected secondary errors, while avoiding detrimental secondary actions whenever preservation is feasible. Therefore, for each EPC, ESC, and EMP combination, the 1-D-LUT stores the SS that is lexicographically optimal with respect to the prioritized error-correction objective. This represents an instantaneous topology-constrained optimum over the admissible SE-VSI SSs, rather than a finite-horizon MPC optimum [4], [5].

To illustrate the SS selection logic, consider four exemplary scenarios corresponding to cases #1 to #4 in Table III. In **Case #1**, the EPC = 100 (EPO = ACB) identifies $\varepsilon_{\text{crit}(a)}$ as the prime error, followed by $\varepsilon_{\text{crit}(c)}$ and $\varepsilon_{\text{crit}(b)}$. The ESC = 011 dictates that i_a must be decreased while i_c and i_b need to be increased if possible. Therefore, the LUT maps the input index to SS = 0110, which decreases i_a , increases i_c , and preserves i_b for a later control cycle. Note that since EMP = 1 (DM predominance), an i_0 -neutral SS is selected, allowing for a double CE correction. In **Case #2**, the EPC is unchanged (ACB), while ESC and EMP change to 001 and 0, respectively, dictating that i_a needs to be decreased, as well as i_c , if possible, while (if possible) i_b needs to be increased. This CM predominance case presents a special scenario where all three CEs can be

corrected simultaneously by selecting the i_0 -impacting SS = 0101, thus decreasing i_0 by the expected 1/3-fold. In **Case #3**, the EPC = 010 (BAC) determines that $\varepsilon_{\text{crit}(b)}$ is the prime error, followed by $\varepsilon_{\text{crit}(a)}$, then $\varepsilon_{\text{crit}(c)}$. The ESC = 110 dictates that i_b must be increased, i_a should also be increased if possible, and i_c should be decreased if possible. Due to EMP = 1, an i_0 -neutral SS is selected, namely SS = 1101, which increases i_b , decreases i_c , and preserves i_a . In **Case #4**, where the LUT input remains the same except for EMP = 0, the selected SS must now contribute to ZSC suppression. Therefore, SS = 1100 is selected. The prime correction of i_b is preserved, i_0 is affected for CM correction, while i_a and i_c are preserved. Thus, compared with **Case #3**, i_c is preserved instead of decreased to satisfy the CM-predominance requirement. This confirms that OEPC dynamically selects SSs based on error priority and their mode predominance by leveraging a 1-D-LUT to efficiently select the optimal SS, ensuring all phase current errors and the ZSC error are always corrected within a few control cycles.

Fundamentally, any control strategy for SE-VSIs, whether 3-D-SVM, CBPWM, or MPC, suppresses ZSC by manipulating the inverter’s CMV to generate a counter-driving force. The distinguishing feature of OEPC is its direct current control architecture. Unlike modulation-based schemes that synthesize an average voltage vector, OEPC directly identifies the instantaneous error priority (e.g., ZSC excursion versus phase current error). When the ZSC error is the most prominent, the LUT selects the specific SS that generates the opposing v_{cm} while simultaneously satisfying the most critical individual phase current requirement. By directly selecting the optimal state per control cycle without modulation or duty-cycle averaging, OEPC achieves the physical goal of ZSC suppression with minimal computational overhead. Furthermore, since the state is only updated when error boundaries are breached (variable switching frequency), unnecessary switching actions are naturally eliminated, reducing switching losses compared with fixed-frequency modulation.

It is important to note that the LUT mapping is derived from the inherent topological constraints of the SE-VSI (Kirchhoff’s laws) rather than parameter-dependent tuning. Therefore, the logic remains valid globally, regardless of load variations or non-ideal parasitics. Furthermore, the memory footprint of 96 entries (approx. 100 bytes) is negligible for modern microcontrollers, presenting no practical implementation drawback compared with the kilobyte-sized code overhead of online methods such as MPCC or 3-D-SVM. It should be clarified that the OEPC’s 1-D-LUT is not a parametric map tuned for specific operating points, but rather an exhaustive logic truth table. The 96 entries represent the complete set of all mathematically possible permutations of error priorities (6 states), error signs (8 states), and mode predominance (2 states).

Consequently, there are no ‘omitted conditions’: at any given instant, the system’s error vector must fall into one of these defined logic states. Furthermore, since the mapping is derived strictly from the SE-VSI topology (Kirchhoff’s laws) rather than motor parameters, the decision logic remains valid globally. Even under non-ideal conditions, such as parameter mismatch, saturation, or fault, the direction of the current response

to a given SS remains invariant. Thus, the LUT guarantees the correct counter-action regardless of the plant's physical non-idealities, as validated by the unbalanced load experiments.

D. Overall OEPC Algorithm and Hardware Implementation Considerations.

The OEPC's overall framework is illustrated as a flowchart in Fig. 3, outlining the main procedural steps from current sensing to gate signal generation. The LUT-based optimized SS selection, which was detailed in Section III-C, is represented by a black-boxed subprocess. The OEPC algorithm depicted in Fig. 3 proceeds through the following main systematic stages.

- 1) *Error Generation*: Six current control errors are calculated—three DM errors (8) and three CM errors (9), representing deviations between reference and measured currents and quantifying all possible phase-wise correction demands.
- 2) *Error-Priority Processing*: Each phase's DM and CM errors are pairwise compared with select the dominant error (10). This reduces the total set to three CEs, one per phase, while simultaneously determining the EMP flag, which signals whether DM or CM correction should dominate the SS selection (11).
- 3) *Error Sorting and Classification*: The three CEs are sorted by magnitude, establishing a hierarchical error priority by generating the EPC (12) and the ESC (13), which encode the CEs magnitudes and signs, respectively. Together with the EMP flag, they form a 7-bit address uniquely identifying the system's error state (14).
- 4) *LUT-Based SS Selection*: The 7-bit address indexes into the 1-D-LUT to determine the optimal SS by implementing the four fundamental rules described in Section III-C, ensuring that: (a) the prime error is always addressed; (b) maximum error correction is achieved where possible; (c) non-correctable errors are preserved through self-circulation; and (d) The balance between phase current regulation and ZSC suppression dynamically adapts based on the EMP.
- 5) *Gate Signal Generation*: The selected SS is decoded into eight complementary gate signals controlling the SE-VSI's four legs, which are refreshed each control cycle, at a self-adjusted f_{sw} with a bounded upper limit.

Since the LUT selection is based solely on instantaneous error priorities and signs, the control logic is independent of the f_{sw} range, which is determined by the chosen sample time. The OEPC loop simply executes the optimal SS selection at whatever rate the hardware permits [up to f_{max} , given by (7)], ensuring consistent performance across different operating frequencies. The sample time should be selected based on hardware limitation, ensuring it remains larger than the maximum execution time. It is also worth noting that the OEPC's self-oscillating nature produces a spread-spectrum profile that distributes harmonic energy across a wider bandwidth than fixed-frequency modulation. The switching frequency remains bounded between $0.5f_{max}$ and f_{max} , averaging at $0.75f_{max}$

(cf. Section IV-A). This constrained high-frequency range reduces conducted EMI while keeping the lowest harmonics high enough for the stator inductance to provide effective filtering without additional resonant traps. It should be indicated that the adaptive switching frequency mechanism is an optional extension enabled by the event-driven nature of OEPC and is not required for its core ZSC suppression capability. Although not explored here for the sake of brevity, a fixed f_{sw} operation is also feasible.

A key feature of OEPC architecture is its computational efficiency, driven by the direct lookup, thus eliminating the iterative calculations typical of 3-D-SVM or predictive controllers. This allows for the support of hardware-efficient implementation. While validated on the TMS320F28335 MCU, the core LUT operation is reducible to a seven-input, four-output truth table. It can be directly mapped onto compact PLAs, such as the GAL22V10. Paired with cost-effective MCUs (e.g., STM32F031) for error computation and speed loop handling, this MCU-PLA hybrid approach offers significant potential for further reducing system cost, latency, and power loss. Realization of the LUT as combinational logic enables fast response times, making OEPC suitable for applications demanding rapid control actions.

IV. VALIDATION AND PERFORMANCE EVALUATION

OEPC was validated through three stages, ensuring rigor, practical applicability.

- 1) *Digital Simulations*: Conducted in MATLAB/Simulink to verify theoretical principles.
- 2) *Real-Time C-HIL*: Implemented on a high-fidelity Typhoon HIL402 platform ($T_{sim} \approx 0.5 \mu s$) to evaluate control performance on an OW-PMSM drive using real target hardware (TI TMS320F28335, $T_s = 25 \mu s$).
- 3) *Experimental Prototype*: Performed on a physical SE-VSI prototype driving an unbalanced RL load to stress-test ZSC suppression capability. Inductance asymmetry was introduced by manually adjusting the core entry depths of the load inductors, creating distinct phase values ($L_a < L_b < L_c$) and a controlled $\pm 25\%$ imbalance.

Digital simulations used a balanced resistor-inductor-EMF (RLE) load, an unbalanced RL load, and an OW-PMSM drive. The PMSM model [16], implemented in Simulink, employed a linear VBR-based approach, enhanced to include third-harmonic ripple effects [cf. (1)–(5)]. The OEPC algorithm served for current control in both cases. For real-time C-HIL validation [Fig. 4(a)], a Typhoon HIL402 emulator [22] executed the plant model with a $0.5 \mu s$ time step to capture $2 \mu s$ deadtime effects. The C-HIL control algorithm ran on a DSP180 interface board [23] with a 150 MHz TI TMS320F28335 DSP [24]. In the laboratory prototype [Fig. 4(b)], the SE-VSI used 650 V GaN transistors (GS66S16B) [25] and 650 V SiC Schottky diodes (IDL12G65C5) [26]. It drove a three-phase physically unbalanced RL load with $\pm 25\%$ inter-phase parameter variations to test robustness. The prototype operated at a 60 V DC link voltage (V_{DC}) and a maximum control frequency (f_{max}) of 20 kHz. The algorithm ran on a standalone 200 MHz

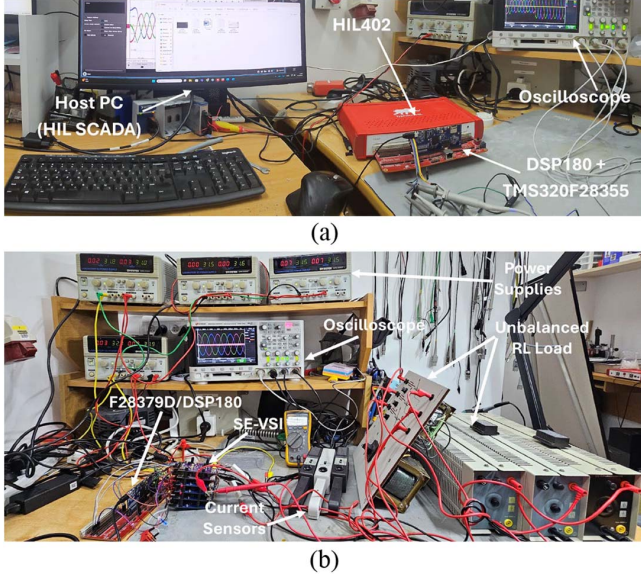


Fig. 4. Experimental setups. (a) Controller + hardware in loop (C-HIL) test rig. (b) Physical prototype test rig.

TI TMS320F28379D DSP with a sampling time of $T_s = 25 \mu s$. A Keysight MSO-X3014T oscilloscope was utilized for signal analysis.

A. OEPC Current Control Tests

While unsupported by OEPC - an inherently current control scheme, equivalent modulation indices of $\alpha\beta$ subspace, $m_{\alpha\beta}$, and CMV, m_{zs} , were computed for consistent comparison with voltage modulation-based methods, such as 3-D-SVM [6] and CBPWM [8]. For the RLE load case, $m_{\alpha\beta}$ was computed from the fundamental current magnitude, I_f , while m_{zs} was set by controlling the 3rd harmonic back-EMF (E_0) as

$$m_{\alpha\beta} = (I_f/V_{DC}) \sqrt{R^2 + (\omega L)^2}, \quad m_{zs} = 3E_0/V_{DC}. \quad (15)$$

The mutual relationship between $m_{\alpha\beta}$ and m_{zs} is governed by the geometric modulation constraint, $m_{\alpha\beta} + (2/3)m_{zs} \leq 1$, [6], which defines the maximum allowable $m_{\alpha\beta}$ as a function of m_{zs} in the linear range, thus capturing the space sharing between DM and CM error correction feasibility in OEPC

$$m_{\alpha\beta_max} = \min \left(1 - \frac{2}{3}m_{zs}, 1 \right). \quad (16)$$

Note that the $\sqrt{3}/2$ scaling factor is applied strictly to $m_{\alpha\beta}$, so that the $\alpha\beta$ -subspace linear range is reported on a normalized 0–1 scale. This normalization is not applied to m_{zs} , which remains defined according to (15) and the convention of [6].

In the first set of tests, OEPC's sinusoidal current tracking, ZSC behavior, f_{sw} variability, and DC utilization capabilities were evaluated under different operating conditions using an RLE load, where E_0 and I_f^* are determined by the desired $m_{\alpha\beta}$ and m_{zs} values according to (15). R , L , V_{DC} , and f_{max} , were set to 8Ω , 5 mH , 60 V , and 5 kHz , respectively, to align with the 3-D-SVM [6] RL load test conditions. Notably, OEPC operates as a self-oscillating scheme with a typical steady-state average f_{sw} , $\langle f_{sw} \rangle$ of approximately $0.75f_{max}$, which

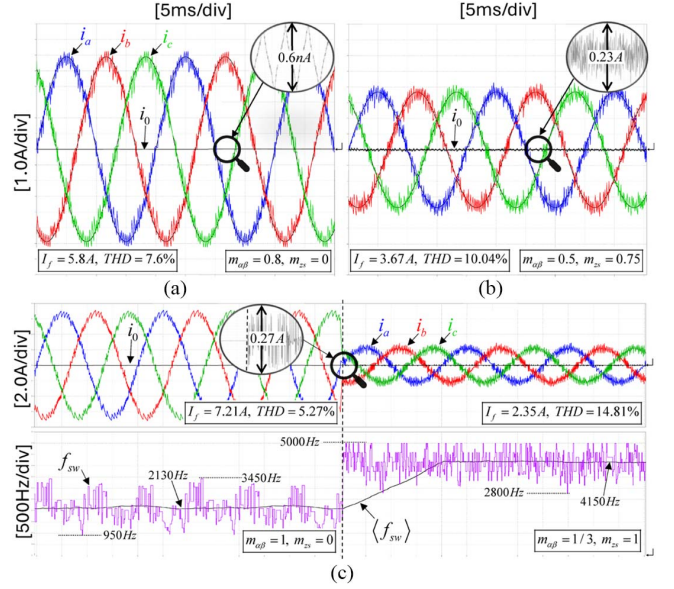


Fig. 5. OEPC sinusoidal current tracking results under different conditions ($f_{max} = 5 \text{ kHz}$): (a) i_x, i_0 ($m_{\alpha\beta} = 0.8, m_{zs} = 0$); (b) i_x, i_0 ($m_{\alpha\beta} = 0.5, m_{zs} = 0.75$); (c) i_x, i_0, f_{sw} ($m_{\alpha\beta} = 1 \rightarrow 1/3, m_{zs} = 0 \rightarrow 1$).

gives fixed- f_{sw} schemes like 3-D-SVM, a potential advantage in current quality performance. Phase currents (i_x), ZSC (i_0) along with I_f and total harmonic distortion (THD) results are illustrated in Figs. 5(a)–(c), while in Fig. 5(c) the cycle-to-cycle f_{sw} and $\langle f_{sw} \rangle$ are also revealed. The three figures depict the results of three test cases.

- 1) Ideal switches, linear load ($m_{\alpha\beta} = 0.8, m_{zs} = 0$): The absolute zero i_0 reveals the OEPC's capability to replicate a three-wire star-connected load behavior under CBPWM.
- 2) Current quality at non-ideal switching ($m_{\alpha\beta} = 0.5, m_{zs} = 0.75$): Despite the advantage of 3-D-SVM with a fixed $f_{sw} = 5 \text{ kHz}$ and a lower m_{zs} of 0.5, at an average f_{sw} of 3.75 kHz , OEPC maintains a competitive THD of 10% and, as quantified in the comparative analysis in Table V, achieves a superior raw i_0 ripple of 0.23 A compared with 1.20 A for 3-D-SVM [6] (under matched conditions). This confirms that even without complex sector modulation, the error-priority logic effectively clamps the ZSC component.
- 3) Transient behavior and maximum DC utilization ($m_{\alpha\beta} = 1 \rightarrow 1/3, m_{zs} = 0 \rightarrow 1$): Simultaneously changing $m_{\alpha\beta}$ and m_{zs} between their boundary limits as per (15) while monitoring f_{sw} variation and $\langle f_{sw} \rangle$, highlights OEPC's robust transient behavior and efficient f_{sw} adaptivity, which reflects a general trait of comparator-based controllers, where lower reference magnitudes lead to higher f_{sw} due to rapid and aggressive correction/reversal cycles causing faster accumulation of error. Moreover, when f_{max} was increased from 5 to 50 kHz for this test, current THD and ZSC ripple were both decreased from 14.81% and 2.35 A to 1.42% and 0.023 A , respectively, while maintaining maximal DC utilization, demonstrating OEPC's full ZSC suppression capability even at saturation

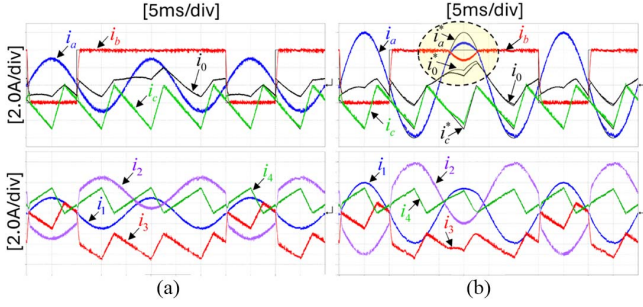


Fig. 6. OEPC response (phase and leg currents) under asymmetric references operation with transient overmodulation. (a) Maintained modulation constraints. (b) Periodically violated modulation constraints.

boundary while maintaining minimal switching losses impact.

A second set of tests was conducted to assess the modulation boundary using three asymmetric reference currents: a sine wave, a rectangular wave, and a triangular wave, each characterized by distinct frequencies, amplitudes, duty cycles, phase shifts, and DC offsets. V_{DC} , L , and R , were consistent with the first test, while E_0 and f_{max} were set to 0 and 10 kHz, respectively. Fig. 6 displays the simulation results, which include the reference (i_x^*), the measured (i_x) currents, the ZSC (i_0), its effective reference (i_0^*), and the leg currents ($i_1 - i_4$). In Fig. 6(a), the 3.0 A sinusoidal reference (i_a^*) remained within the modulation constraints, enabling accurate current tracking.

In Fig. 6(b), doubling i_x^* to 6.0 A caused it to exceed the boundary periodically. During those overmodulation intervals, OEPC displayed unique self-regulating behavior. Instead of saturating, it dynamically distributed control priority among the competing errors, causing temporary current underflows and full recovery once the feasible modulation was restored. These results illustrate OEPC's robustness under extreme asymmetry and its ability to handle transient infeasibility gracefully. While conventional overmodulation support could be incorporated for applications requiring saturation (e.g., via logic flags or LUT switching), OEPC's default behavior inherently prevents harmonic distortion and maintains control predictability.

To extensively validate the proposed method's ZSC suppression capability independent of motor back-EMF harmonics, a series of static RL load tests were conducted. These tests were designed to match the specific constraints of the laboratory prototype to ensure a direct 1-to-1 comparison between simulation and experiment. The system parameters were set to $R = 8 \Omega$, $L = 5 \text{ mH}$, and $V_{DC} = 60 \text{ V}$, with a control frequency bound of $f_{max} = 20 \text{ kHz}$. For these tests, the DM modulation index ($m_{\alpha\beta}$) was strictly limited to 0.8 (peak phase current $\approx 6 \text{ A}$). Such derating was imposed for two reasons.

- 1) *Theoretical Constraint*: As defined by the reciprocal of (16), $m_{zs_min} = \min(3/2(1 - m_{\alpha\beta}), 1)$.
- 2) Setting $m_{\alpha\beta} = 0.8$ reserves sufficient voltage headroom ($m_{zs} = 0.3$) for the controller to inject the necessary zero-sequence voltage required to balance the currents against the $\pm 25\%$ impedance asymmetry.

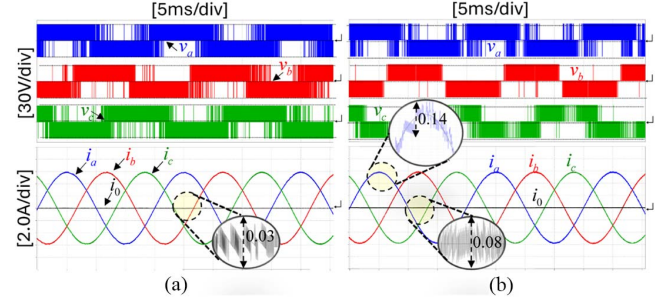


Fig. 7. OEPC current control simulation results with RL load (phase leg to leg voltages and currents) at $V_{DC} = 60 \text{ V}$, $f_{max} = 20 \text{ kHz}$, and $m_{\alpha\beta} = 0.8$. (a) Balanced load. (b) Physically unbalanced load with $\pm 25\%$ arbitrary R , L variations.

- 3) *Hardware Constraint*: The laboratory DC power supply configuration limited the maximum sustainable peak current to 6 A per phase.

Fig. 7 presents the simulation results for both balanced and unbalanced conditions. In the balanced case [Fig. 7(a)], the ZSC is virtually eliminated, exhibiting only a negligible ripple of 0.03 A attributed to the deadtime effects. The phase currents remain perfectly sinusoidal with a THD of just 0.8%.

In the unbalanced case [Fig. 7(b)], severe physical asymmetry was introduced by varying the phase impedances by $\pm 25\%$. (Phase A: -25% , Phase B: Nominal, Phase C: $+25\%$). Despite this significant mismatch, the OEPC logic successfully maintained balanced phase currents with a virtually unaffected THD of 0.84%. The ZSC ripple remained tightly clamped at 0.8 A, confirming the controller's robustness in identifying and suppressing common-mode errors even when the plant is physically asymmetric.

To experimentally verify these findings, a custom SE-VSI prototype was constructed and tested under the same unbalanced conditions shown in Fig. 7(b). The prototype, shown in Fig. 4(b), utilizes GaN Systems GS66516B (650 V, 60 A) enhancement-mode transistors paired with Infineon IDL12G65C5 SiC Schottky freewheeling diodes. The control algorithm executes on a modern TI TMS320F28379D dual-core DSP (200 MHz). To maximize hardware efficiency on this platform, the OEPC algorithm was optimized by replacing standard floating-point modulo operations with bitwise integer logic and eliminating the default LUT interpolation. This optimization reduced the core current control execution time to just $0.85 \mu\text{s}$, demonstrating the method's suitability for high-frequency industrial control loops.

Figs. 8 and 9 show experimental steady-state waveforms from the prototype under the same $\pm 25\%$ unbalanced load condition used in simulation. Fig. 8. illustrates representative differential phase voltages (v_a and v_c), measured between SE-VSI Legs 1–2 and Legs 3–4, respectively. The waveforms highlight the distinct duty cycles applied to Phases A and C to compensate for their impedance mismatch ($Z_a < Z_{nom} < Z_c$), where the effective modulation depth of v_c exceeds that of v_a (v_b is omitted due to oscilloscope limitations). Fig. 9 presents the resulting three-phase currents (i_a, i_b, i_c) and ZSC (i_0), computed in real time via the oscilloscope's math function. Consistent

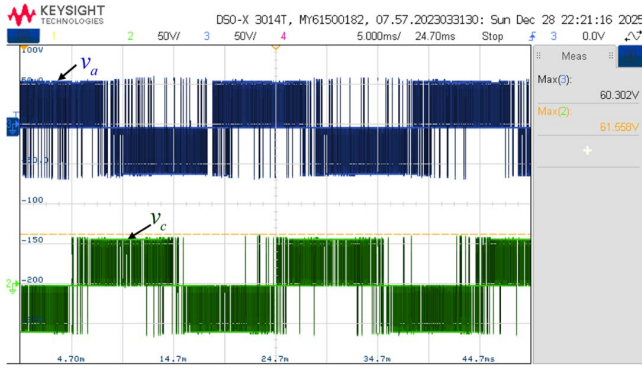


Fig. 8. Experimental differential phase voltages (v_a and v_c), measured across SE-VSI legs 1–2 and 3–4, respectively, illustrating the distinct duty cycles required to balance a $\pm 25\%$ physically unbalanced RL load ($V_{DC} = 60$ V, $f_{max} = 20$ kHz).

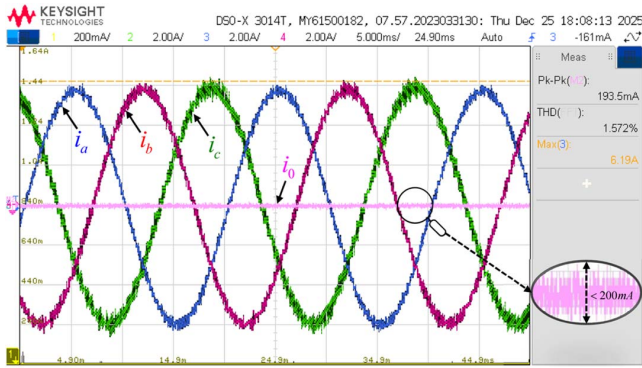


Fig. 9. Experimental three-phase currents (i_a, i_b, i_c) and computed zero-sequence current (i_0 , via oscilloscope math) under $\pm 25\%$ parameter asymmetry, illustrating effective ZSC suppression and phase balancing ($V_{DC} = 60$ V, $f_{max} = 20$ kHz).

with simulation, the experimental currents remain balanced and sinusoidal despite severe impedance asymmetry. Fast Fourier transform (FFT) analysis shows a phase current THD of 1.57% and a peak-to-peak ZSC ripple of 0.19 A, slightly exceeding simulation values (0.84% THD, 0.08 A ripple) due to measurement noise and switching parasitics, confirming the control's high fidelity. Balanced load tests (omitted for brevity) achieved 1.08% THD and 0.15 A ZSC ripple. Comparing both scenarios confirms effective ZSC suppression with only a marginal THD increase, validating trends observed in simulation.

Across all tests, the modulation range and DC-bus utilization complied with constraints in (15) and (16), confirming OEPC's adherence to SE-VSI limits while ensuring robust current control and high quality. By dynamically exploiting the full voltage vector space under error-priority logic, OEPC achieves maximal DC-bus utilization inherent to the topology, avoiding suboptimal constraints imposed by fixed carrier-based dead-time compensation.

B. OEPC Motor Drive Tests

To ensure high-fidelity modeling of the inverter switching events and deadtime effects ($\approx 2 \mu s$), the HIL plant emulation (SE-VSI and SEW-PMSM) was executed on the Typhoon

TABLE IV
OW-PMSM TEST CONFIGURATIONS (PMSM PARAMETERS)

#	V_{DC} [V]	L_d, L_q, L_0 [mH]	R_S [Ω]	p	J [gm^2]	λ_m [mWb]	K_m [%]
1	48	3.7, 5.0, 4.0	0.9	5	1.0	80	2.5
2	60	1.5, 1.8, 2.2	0.4	5	1.0	80	2.5
3	60	3.7, 5.0, 4.0	0.9	5	1.0	22	2.5
4	200	4.0, 7.0, 6.0	2.0	4	2.0	175	5.0

TABLE V
REPORTED PMSM DRIVE RESULTS VERSUS
OEPC-BASED REPLICATIONS

Methodology	Test Cfg #	Speed [rpm]	T_m [Nm]	I_f [A]	Δi_0 [A]	THD [%]
MVMPC [4]	1	100	0.4	1.5	0.20	6.10
OEPC					0.17	6.84
MVMPC [4]	1	1000	2.3	8.0	1.60	7.51
OEPC					0.24	1.69
AZSMPC [5]	2	100	0.5	3.0	0.18	5.02
OEPC					0.14	5.75
AZSMPC [5]	2	500	0.5	3.1	0.20	5.81
OEPC					0.14	6.54
3-D-SVM [6]	3	500	1.0	3.8	1.05	-
OEPC					0.24	4.31
3-D-SVM [6]	3	1000	1.5	5.0	1.20	-
OEPC					0.27	3.08
CBPWM [8]	4	1000	15	40	1.57	3.12
OEPC					0.49	2.26

Note: Bold entries indicate the optimal performance metrics and superior results achieved by the proposed method compared to conventional control strategies.

HIL402 platform [22]. The solver was configured with the 'Exact' discretization method, and generic digital signal (GDS) oversampling enabled, utilizing an effective simulation time step of $T_{sim} \approx 0.5 \mu s$.

In contrast, the control algorithm runs on the physical TMS320F28335 DSP host card. The control code, generated via PLECS and deployed via TI code composer studio (CCS), executes within an interrupt service routine (ISR) triggered at the variable switching frequency rate. As per (7), for the configured $f_{max} = 20$ kHz, the control sampling time is set to $T_s = 25 \mu s$. This separation ensures that the validation captures the realistic impact of digital discretization and processor latency on the high-fidelity plant model.

The OEPC-based SEW-PMSM tests were conducted across four configurations, which are detailed in Table IV. Configurations #1–#3 fully align with those from [4], [5], and [6], respectively, while #4 is a higher-power OW-PMSM drive with a pronounced 3rd harmonic flux component. Table V lists the reported results from [4], [5], [6] at $f_{max} = 20$ kHz, alongside OEPC results under identical conditions (superior results are bolded). Table V reveals that OEPC improves ZSC ripples and achieves competitive current THD.

It should be noted that discrepancies were observed between reported torque levels and implied current magnitudes in [4] and [6], where results could not be replicated using the stated machine models and test conditions unless a field weakening condition of $i_d^* = -2i_q^*$ was enforced, deviating from the typical $i_d^* = 0$ in standard FOC [14]. This was the only way to align simulation results with the reported 2x fold inflated current

TABLE VI
COMPUTATIONAL BURDEN COMPARISON OF CONTROL METHODS

Method	Platform	Cores	Clock Speed	Execution Time	Cost
OEPC	TMS320F28335	1	150 MHz	1.75 μ s	100%
3-D-SVM [6]	dSPACE DS1202	2	2.0 GHz	0.43 μ s	330%
CBPWM [8]	dSPACE DS1202	2	2.0 GHz	2.10 μ s	1600%
V ³ -SVM (Data from [6])	dSPACE DS1202	2	2.0 GHz	2.50 μ s	1900%
MV-MPCC [4]	dSPACE DS1202	2	2.0 GHz	>20 μ s	15000%

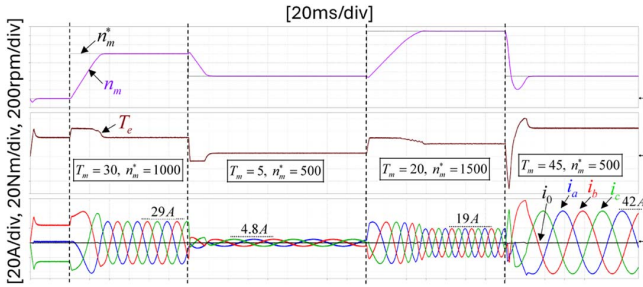


Fig. 10. Dynamic response of the SEW-OEPC-based PMSM drive under speed and load variations.

amplitudes in [4] and [6] (i.e., 8 A at 2.3 Nm). While this behavior may stem from machine-related constraints or reflect a design choice, it was not explicitly addressed in [4] and [6].

Fig. 10 presents the dynamic response of the SEW-OEPC-driven OW-PMSM drive, corresponding to configuration #4 in Table IV [240 volts alternating current (VAC), 30 Nm, 1000 revolutions per minute (rpm) rated] at $f_{\max} = 20$ kHz, under sequential speed and load torque variations. The test sequence progresses through multiple steady-state operating points (T_m , n_m) including low-speed light-load (500 rpm, 5 Nm), high-speed medium-load (1500 rpm, 20 Nm), and abrupt transitions such as acceleration from standstill to 1000 rpm at full load and deceleration from 1500 to 500 rpm while simultaneously increasing the load torque to 45 Nm. Throughout these transitions, the drive demonstrates excellent reference tracking, disturbance rejection, and current regulations. Torque ripple remains within ± 0.5 Nm under all conditions, with no current transient oscillations observed, highlighting the OEPC's robust control performance.

Fig. 11 compares the steady-state responses of SE-VSI-driven OW-PMSM (configuration #4) using OEPC and CBPWM [8] under prominent 3rd harmonic back-EMF conditions, with K_m increased to 20% at rated speed and 50% load ($n_m = 1000$ RPM, $T_m = 15$ Nm), resulting in modulation indices of $m_{\alpha\beta} = 0.62$ and $m_{zs} = 0.66$, leading to 6% overmodulation ($m_{\alpha\beta} + 2/3m_{zs} = 1.06$), as per (16). While such a high K_m is impractical (typically $<5\%$), it ensures quite a challenging scenario for ZSC and torque ripple suppression. As Fig. 11(a) shows, OEPC ($f_{\max} = 20$ kHz) achieves low torque and ZSC ripples of 2.2 Nm and 0.49 A with 2.26% phase current THD. Conversely, the CBPWM results

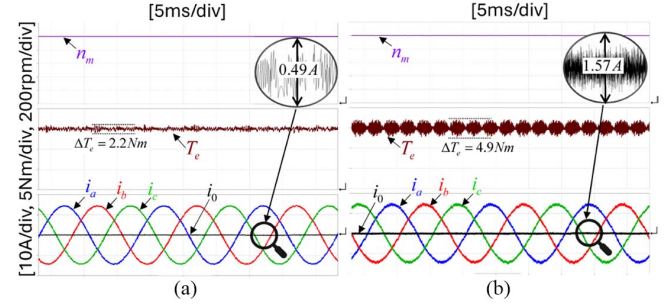


Fig. 11. Steady-state speed, torque, and phase current responses of SE-VSI-driven OW-PMSM under high 3rd harmonic flux conditions ($K_m = 20\%$) and 6% overmodulation. (a) OEPC. (b) Benchmark CBPWM [8].

in Fig. 11(b) show higher ripples (4.9 Nm, 1.57 A) and 3.12% current THD. CBPWM also exhibits a strong 6th harmonic component in torque ripple from its linear ZSC compensator, causing degraded performance near overmodulation [17]. Despite operating at lower f_{sw} ($f_{sw} \approx 16$ kHz), OEPC reduces switching losses by $\sim 20\%$ compared with CBPWM, thanks to its adaptive f_{sw} utilization. These results confirm OEPC's superior ZSC and torque ripple suppression, improved current quality, and reduced switching losses even under harsh conditions.

Fig. 12 presents real-time C-HIL experimental results of OEPC-based motor drive control using the same motor configuration #4 and $f_{\max} = 20$ kHz limit as in the simulations. The test employs an emulated OW-PMSM with a 10-bit pulses per revolution (PPR) and a 20 kHz BiSS-based resolver carrier, representative of real-world applications. Unlike the complex trajectory stress-test used in the simulation (Fig. 10), the C-HIL validation utilizes a sequential step-response profile to isolate and clearly characterize the dynamic performance of the speed and current loops.

Starting at 500 rpm/15 Nm, the speed reference was stepped to 1000 rpm, followed by a load torque increase to 30 Nm (via the HIL supervisory control and data acquisition (SCADA) interface), while capturing the triggered events' responses. Fig. 12(a) shows the speed and torque responses, while Fig. 12(b) shows the phase currents and ZSC, all demonstrating excellent dynamic performance and regulation, validating the simulation predictions. In addition, complete control execution time measured from ADC trigger of speed sensor (encoder) to general-purpose input/output (GPIO) update at 40 kHz sampling ($f_{\max} = 20$ kHz) revealed that the OEPC cycle, including FOC, completed in 1.75 μ s on the TMS320F28335 (150 million instructions per second (MIPS), single core). To further contextualize this performance, Table IV presents a comparative analysis of the computational burden against state-of-the-art methods reported in [4], [6], and [8]. Note that direct execution time comparisons are skewed by the hardware platforms: recent methods typically utilize high-performance, dual-core dSPACE DS1202 systems (2.0 GHz), whereas OEPC is implemented on a standard, single-core TI TMS320F28335 (150 MHz). To account for this, the normalized Cost (%) column standardizes the

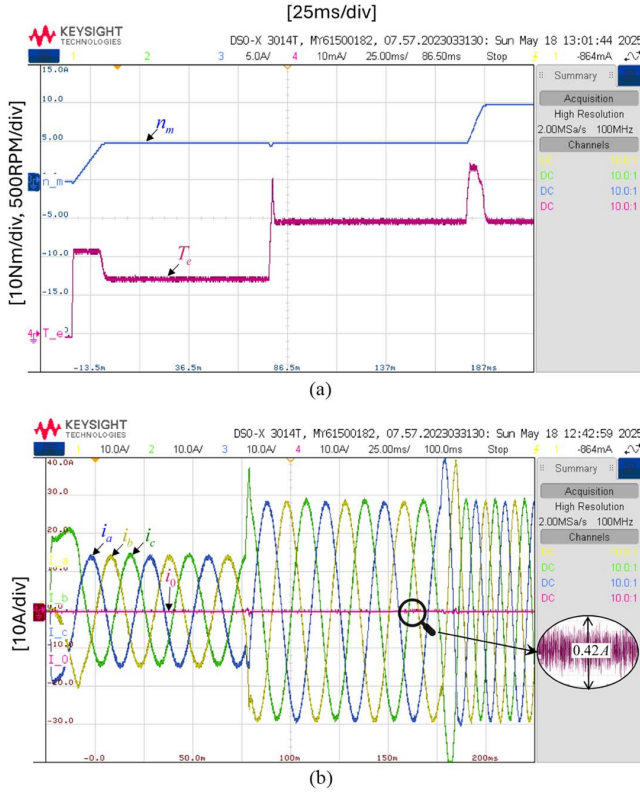


Fig. 12. Real-time experimental results of OEPC-based motor drive control (C-HIL with TMS320F28335 MCU) and emulated OW-PMSM. (a) Speed and electrical torque response under sequential speed and load changes. (b) Current (phase + ZSC) responses.

computational load in terms of processor clock cycles relative to OEPC. The results reveal that while 3-D-SVM [6] achieves a low absolute time ($0.43 \mu\text{s}$) driven by the 2.0 GHz clock, its algorithmic complexity is 3.3 times higher than OEPC. Furthermore, standard methods like CBPWM [8] and Virtual-Vector SVM (V^3 -SVM) exhibit computational loads 16 to 19 times higher per core. This confirms that OEPC is algorithmically superior, capable of delivering high-performance control on cost-effective, single-core industrial microcontrollers (Note: On the newer TMS320F28379D platform used for the prototype, the optimized OEPC current loop executed in just $0.85 \mu\text{s}$, indicating further potential for latency reduction).

At steady-state full load ($n_m = 1000 \text{ RPM}$, $T_m = 30 \text{ Nm}$), the measured power efficiency for the SE-VSI reached 94% when accounting for parasitics, semiconductor losses, and deadtime. FFT analysis revealed a current THD of 1.84% and a 0.42 A ZSC ripple, closely matching simulated results. These findings confirm OEPC's practical viability, robustness, and efficiency on cost-effective digital hardware.

C. Intrinsic Fault Robustness Under ITSC Conditions

To rigorously evaluate the controller's robustness under severe internal machine faults, a comparative simulation of an inter-turn short circuit (ITSC) was performed on Phase A of the PMSM (Configuration #4). The fault was modeled using the VBR approach [10], [16] with high fidelity. Phase

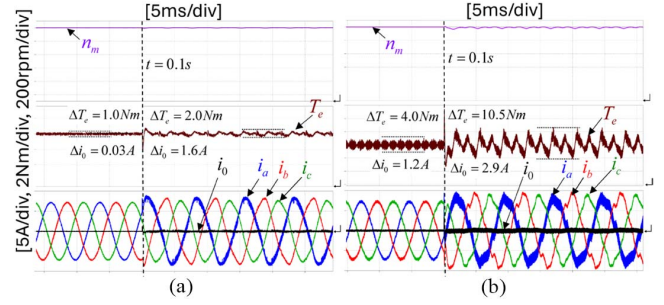


Fig. 13. Comparative dynamic response under a 10% inter-turn short circuit (ITSC) fault on phase A (triggered at $t = 0.1 \text{ s}$, $r_f = 0.1 \Omega$). (a) Proposed OEPC. (b) Benchmark CBPWM.

impedance parameters were divided based on a fault ratio: $k_{sc} = 0.1$ (10% shorted turns). Notably, this represents the maximum fault severity that OEPC can manage at a 15 Nm/100 rpm load and $f_{\text{max}} = 20 \text{ kHz}$ without losing current control. Fig. 13 compares OEPC (a) and CBPWM (b) under identical conditions. As shown in Fig. 13(b), CBPWM fails to contain fault dynamics, resulting in severe distortion: current THD of 11.57%, ZSC ripple of 2.9 A, and torque ripple reaching 10.5 Nm.

In contrast, Fig. 13(a) demonstrates the superior stability of OEPC. At fault onset, the switching frequency reaches its maximum limit, producing visible PWM jitter on Phase A due to intensified switching. This rapid response enables OEPC to clamp i_0 to 1.6 A and restrict torque ripple to approximately 2 Nm. The resulting THD of 6.76% must be interpreted within this severe postfault survival regime rather than normal operation. In the OEPC framework, current distortion depends strongly on modulation behavior, switching activity, and load dynamics. Under healthy conditions, distortion remains exceptionally low (0.8% in simulation, 1.57% in RL experiments, and 1.84% under nominal steady-state motor operation), consistently outperforming conventional carrier-based methods.

Under the severe 10% ITSC fault considered here, the controller prioritizes rapid suppression of large tracking deviations to prevent current runaway. This necessitates simultaneous, non-sequential switching of multiple inverter legs to contain the disturbance. While this aggressive correction ensures effective ZSC suppression, it introduces a moderate harmonic penalty, yielding the observed 6.76% THD. Importantly, this distortion level is not intrinsic to OEPC and can be reduced below the 5% regulatory threshold by tightening error-priority bounds or increasing the controller sampling frequency (T_s). However, such adjustments increase switching activity and associated semiconductor losses, reflecting a standard trade-off between harmonic performance and thermal stress. The reported operating point therefore represents a balanced design choice.

These results demonstrate OEPC's inherent passive fault tolerance without requiring complex active isolation or reconfiguration schemes. It is important to note that OEPC does not rely on explicit fault diagnosis; instead, its constrained switching logic inherently limits current divergence under severe internal faults, providing improved intrinsic system stability.

V. CONCLUSION

This article introduced the OEPC framework for series-end VSI-fed PMSM drives, enabling direct ZSC suppression in ABC coordinates without modulation synthesis. By leveraging a compact 1-D LUT and a variable-frequency phase-wise approach, OEPC achieves efficient ZSC control, reduced switching losses, and high current quality. Comprehensive validation through simulations, real-time C-HIL, and a physical SE-VSI prototype confirmed the controller's robustness, including its ability to maintain balanced currents under severe load asymmetry ($\pm 25\%$) introduced by variable inductor core adjustments.

Comparative fault analysis further demonstrates OEPC's inherent passive tolerance to inter-turn short-circuit faults, effectively clamping the ZSC and preventing the current runaway observed in carrier-based schemes, while maintaining a balanced trade-off between postfault harmonic distortion and power-device thermal reliability.

Notably, OEPC delivers exceptional computational efficiency, achieving sub-2 μ s execution on the TMS320F28335 and sub-1 μ s on the TMS320F28379D, confirming its suitability as a low-overhead alternative to predictive or modulation-based methods.

While the instantaneous lexicographic optimization logic limits long-horizon foresight compared with MPC, it significantly reduces hardware cost and complexity. The framework is adaptable to other converter topologies, such as multiphase SE-VSIs, provided a topology-specific Switching State Impact Table is derived. Future work will focus on extending the LUT approach to multiphase machines and integrating predictive error trends to further enhance dynamic performance.

ACKNOWLEDGMENT

The authors gratefully acknowledge Typhoon HIL Inc. for providing the C-HIL simulation platform.

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